# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

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|---|---|---|---|
| **Course Code:** | CSA1003 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop** |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO1 Statement:** | *Apply advanced methodologies and agile project management techniques to manage software projects effectively* | | |
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| **Date:** | 27/07/2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I* **Y. Ramadevi – 192373065** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Y. Ramadevi.**

## Annexure A – Agile Sprint Planning & User Story Workshop

# Project Title: Smart Library Book Reservation and Management System

## 1. Identify the Project Vision and Stakeholders

### 1.1 Project Vision

The Smart Library Book Reservation and Management System aims to provide a modern, user-friendly, and efficient digital platform for managing library operations in educational institutions. The system enables students and faculty members to search, reserve, borrow, and return books online while allowing librarians to manage inventory, reservations, fines, and reports efficiently. By automating routine library processes, the system improves accessibility, reduces manual work, enhances user satisfaction, and ensures better utilization of library resources.

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### 1.2 Project Objectives

The main objectives of the Smart Library Book Reservation and Management System are:

- Develop a centralized digital library management platform.
- Enable students and faculty members to search books quickly using title, author, category, or ISBN.
- Allow users to reserve available books online before visiting the library.
- Provide real-time information on book availability and reservation status.
- Automate book issue and return processes.
- Automatically calculate overdue fines and maintain payment records.
- Send notifications for reservation confirmations, due dates, overdue books, and book availability.
- Enable librarians to efficiently manage books, members, reservations, and reports.
- Generate reports on issued books, returned books, overdue books, and library usage.
- Improve operational efficiency, reduce paperwork, and enhance the overall library experience.

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### 1.3 Key Stakeholders

The successful implementation of the Smart Library Book Reservation and Management System involves multiple stakeholders, each with specific responsibilities.

| Stakeholder | Role | Responsibilities |
|---|---|---|
| Students | Primary User | Search books, reserve books, borrow and return books, view borrowing history, pay fines, receive notifications. |
| Faculty Members | Primary User | Search, reserve, borrow, and return academic resources, access borrowing history, receive notifications. |
| Librarian | System Administrator | Manage books, approve reservations, issue and receive books, calculate fines, update inventory, generate reports. |
| Library Manager | Decision Maker | Monitor library performance, manage policies, analyze reports, oversee library operations. |
| College Administration | Project Sponsor | Approve project requirements, allocate budget, monitor implementation, ensure compliance with institutional policies. |
| Software Development Team | Development Team | Design, develop, test, deploy, and maintain the application using Agile methodology. |

| | | |
|---|---|---|
| System Administrator (IT Department) | Technical Support | Manage servers, databases, user accounts, backups, security, and system maintenance. |
| Payment Gateway (Optional) | External Service | Process online fine payments securely and provide payment confirmations. |

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## 2. Create Epics and User Stories

### 2.1 What are Epics and User Stories?

In Agile Software Development, an **Epic** is a large functional requirement that can be divided into smaller user stories. A **User Story** is a short, simple description of a feature from the perspective of the end user. User stories help the development team understand user needs and prioritize features during sprint planning.

**User Story Format:**

**As a , I want , so that .**

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### 2.2 Epics

The Smart Library Book Reservation and Management System consists of the following major epics:

| Epic ID | Epic Name | Description |
|---|---|---|
| **EP-01** | User Authentication | Manage user registration, login, and profile management. |
| **EP-02** | Book Search & Reservation | Search books and reserve available books online. |
| **EP-03** | Borrowing & Returning Books | Manage issuing, returning, and renewal of books. |
| **EP-04** | Library Inventory Management | Add, update, delete, and categorize books. |
| **EP-05** | Notifications & Alerts | Notify users about reservations, due dates, and overdue books. |
| **EP-06** | Fine & Payment Management | Calculate overdue fines and manage online payments. |
| **EP-07** | Reports & Analytics | Generate reports for librarians and administrators. |

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### 2.3 User Stories

#### Epic 1 – User Authentication

| User Story ID | User Story |
|---|---|
| **US-01** | As a **student**, I want to register an account, so that I can access library services online. |
| **US-02** | As a **user**, I want to log in securely, so that my personal information is protected. |
| **US-03** | As a **user**, I want to reset my password, so that I can regain access if I forget it. |
| **US-04** | As a **librarian**, I want to manage user accounts, so that authorized users can access the system. |

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#### Epic 2 – Book Search & Reservation

| User Story ID | User Story |
|---|---|
| **US-05** | As a **student**, I want to search books by title, author, ISBN, or category, so that I can quickly find the required book. |

| | |
|---|---|
| **US-06** | As a **student**, I want to view real-time book availability, so that I know whether the book can be borrowed. |
| **US-07** | As a **student**, I want to reserve a book online, so that it is available when I visit the library. |
| **US-08** | As a **faculty member**, I want to cancel a reservation, so that I can free the book for other users if I no longer need it. |

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**Epic 3 – Borrowing & Returning Books**

| User Story ID | User Story |
|---|---|
| **US-09** | As a **librarian**, I want to issue books digitally, so that borrowing records are updated automatically. |
| **US-10** | As a **librarian**, I want to record returned books, so that inventory is updated immediately. |
| **US-11** | As a **student**, I want to renew borrowed books, so that I can extend the borrowing period when permitted. |
| **US-12** | As a **student**, I want to view my borrowing history, so that I can track all previously borrowed books. |

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**Epic 4 – Library Inventory Management**

| User Story ID | User Story |
|---|---|
| **US-13** | As a **librarian**, I want to add new books, so that the library collection remains up to date. |
| **US-14** | As a **librarian**, I want to update book details, so that information remains accurate. |
| **US-15** | As a **librarian**, I want to remove damaged or outdated books, so that the catalogue remains relevant. |
| **US-16** | As a **librarian**, I want to categorize books by subject and genre, so that users can browse efficiently. |

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## 3. Define Acceptance Criteria

### 3.1 Introduction

Acceptance Criteria specify the conditions that a user story must satisfy before it is considered complete. They ensure that the developed functionality meets user expectations and can be tested objectively. In Agile projects, the **Given–When–Then** format is widely used because it clearly defines the system behavior under specific conditions.

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**Acceptance Criteria for Selected User Stories**

**User Story US-01**

**User Story:**
**As a student, I want to register an account, so that I can access library services online.**

| Given | When | Then |
|---|---|---|
| The student is on the registration page | The student enters valid details and clicks **Register** | The system creates a new account and displays a successful registration message. |
| The email already exists | The student submits the registration form | The system displays an error message indicating that the email is already registered. |

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**User Story US-02**

**User Story:**
**As a user, I want to log in securely, so that my personal information is protected.**

| Given | When | Then |
|---|---|---|
| The user has a valid account | The user enters the correct username and password | The system logs the user in and displays the dashboard. |
| The user enters incorrect credentials | The user clicks **Login** | The system displays an "Invalid Username or Password" message. |

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**User Story US-05**

**User Story:**
**As a student, I want to search books by title, author, ISBN, or category, so that I can quickly find the required book.**

| Given | When | Then |
|---|---|---|
| The user is logged in | The user enters a search keyword | The system displays matching books. |
| No matching book exists | The search is performed | The system displays "No books found." |

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**User Story US-07**

**User Story:**
**As a student, I want to reserve a book online, so that it is available when I visit the library.**

| Given | When | Then |
|---|---|---|
| The selected book is available | The user clicks **Reserve Book** | The reservation is confirmed and added to the user's account. |
| The selected book is unavailable | The user attempts to reserve the book | The system informs the user that the book is currently unavailable. |

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**User Story US-09**

**User Story:**
**As a librarian, I want to issue books digitally, so that borrowing records are updated automatically.**

| Given | When | Then |
|---|---|---|
| The student has a valid reservation | The librarian clicks **Issue Book** | The system updates the borrowing record and reduces available stock. |

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**User Story US-10**

**User Story:**
**As a librarian, I want to record returned books, so that inventory is updated immediately.**

| Given | When | Then |
|---|---|---|

| | | |
|---|---|---|
| The borrowed book is returned | The librarian records the return | The system updates inventory and borrowing history. |

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**User Story US-18**

**User Story:**
**As a student, I want to receive due-date reminders, so that I return books on time.**

| Given | When | Then |
|---|---|---|
| A book is due within two days | The reminder schedule runs | The system sends an email or SMS reminder to the user. |

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**User Story US-21**

**User Story:**
**As a librarian, I want the system to calculate overdue fines automatically, so that manual calculations are eliminated.**

| Given | When | Then |
|---|---|---|
| A book is returned after the due date | The return is recorded | The system calculates and displays the overdue fine automatically. |

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## 4. Prioritize the Product Backlog

### 4.1 Product Backlog

The Product Backlog is a prioritized list of features required to develop the Smart Library Book Reservation and Management System. Prioritization is based on **Business Value**, **Customer Needs**, and **Technical Dependencies**.

| Priority | User Story ID | Feature | Business Value |
|---|---|---|---|
| **High** | US-01 | User Registration | Very High |
| **High** | US-02 | User Login | Very High |
| **High** | US-05 | Search Books | Very High |
| **High** | US-06 | View Book Availability | High |
| **High** | US-07 | Reserve Books | Very High |
| **High** | US-09 | Issue Books | Very High |
| **High** | US-10 | Return Books | Very High |
| **High** | US-13 | Add Books | High |
| **Medium** | US-11 | Renew Books | Medium |
| **Medium** | US-12 | Borrowing History | Medium |
| **Medium** | US-17 | Reservation Notification | Medium |
| **Medium** | US-18 | Due Date Reminder | Medium |
| **Medium** | US-21 | Fine Calculation | Medium |
| **Low** | US-23 | Online Fine Payment | Low |
| **Low** | US-24 | Borrowing Reports | Low |
| **Low** | US-26 | Popular Book Analytics | Low |

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**Product Backlog Priority Chart**

| Priority | Number of Features |
| --- | --- |
| ⚫ High | 8 |
| ▯ Medium | 5 |
| ▯ Low | 3 |

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## 5. Plan the Sprint Backlog

### Sprint Information

| Item | Details |
| --- | --- |
| Sprint Number | Sprint 1 |
| Sprint Duration | 2 Weeks |
| Sprint Goal | Develop the core library management functionalities required for user authentication and book reservation. |

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### Sprint Backlog

| User Story | Development Tasks |
| --- | --- |
| US-01 User Registration | Design Registration UI, Create Registration Database Table, Develop Backend API, Validate User Input, Unit Testing |
| US-02 User Login | Design Login Page, Implement Authentication, Session Management, Password Encryption, Testing |
| US-05 Search Books | Design Search Interface, Develop Search API, Connect Database, Test Search Results |
| US-06 View Book Availability | Create Availability Module, Database Integration, UI Development |
| US-07 Reserve Book | Reservation Module, Database Update, Notification Integration, Testing |
| US-13 Add Books | Develop Admin Interface, CRUD Operations, Database Testing |

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### Sprint Task Board

| To Do | In Progress | Testing | Done |
| --- | --- | --- | --- |
| Registration Module | Login Module | Search Module | Project Setup |
| Book Search | Reservation Module | Add Book Module | Database Design |
| Book Availability | | | |

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## 6. Estimate Effort Using Story Points

### 6.1 Estimation Technique

The project uses the **Fibonacci Story Point Estimation** technique:

**Story Points:** 1, 2, 3, 5, 8, 13, 21

Story points represent the relative effort required to implement a user story by considering complexity, risk, and development time.

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**Story Point Estimation Table**

| User Story | Story Points | Justification |
|---|---|---|
| **US-01 User Registration** | 5 | Moderate validation, database integration, and testing. |
| **US-02 User Login** | 5 | Authentication, session management, and security implementation. |
| **US-05 Search Books** | 8 | Search algorithms, filtering, and database queries. |
| **US-06 View Book Availability** | 3 | Simple database retrieval and display. |
| **US-07 Reserve Book** | 8 | Reservation logic, validation, inventory update, and notifications. |
| **US-13 Add Books** | 5 | CRUD operations with database integration. |

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**Total Story Points**

| Sprint | Total Story Points |
|---|---|
| **Sprint 1** | **34 Story Points** |

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**Justification**

- **5 Story Points** – Moderate complexity requiring form validation and database integration.
- **8 Story Points** – High complexity involving multiple modules, business logic, and testing.
- **3 Story Points** – Simple functionality with minimal development effort.

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## 7. Present the Sprint Goal and Expected Deliverables

### 7.1 Sprint Goal

The primary goal of **Sprint 1** is to develop the core functionalities of the **Smart Library Book Reservation and Management System**, enabling users to securely register, log in, search for books, view book availability, reserve books, and allow librarians to add books to the library catalog.

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### 7.2 Sprint Duration

| Sprint | Duration |
|---|---|
| Sprint 1 | 2 Weeks |

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### 7.3 Expected Sprint Deliverables

At the end of Sprint 1, the following features will be fully implemented:

- ✓ User Registration Module
- ✓ Secure User Login
- ✓ Book Search Functionality

- ✓ Real-Time Book Availability Display
- ✓ Online Book Reservation
- ✓ Book Management (Add Books)
- ✓ Database Integration
- ✓ User Authentication
- ✓ Unit Testing
- ✓ Sprint Review Demonstration

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**Sprint Deliverable Summary**

| **Deliverable** | **Status** |
|---|---|
| Registration Module | ✓ Completed |
| Login Module | ✓ Completed |
| Search Books Module | ✓ Completed |
| Book Availability Module | ✓ Completed |
| Reservation Module | ✓ Completed |
| Add Book Module | ✓ Completed |
| Database Integration | ✓ Completed |
| Unit Testing | ✓ Completed |

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## Conclusion

The Agile Sprint Planning process for the **Smart Library Book Reservation and Management System** establishes a clear roadmap for development. The project requirements have been organized into prioritized user stories with measurable acceptance criteria. High-priority features have been selected for the first sprint, estimated using the Fibonacci story point technique, and broken down into actionable development tasks. With a well-defined sprint goal and realistic deliverables, the development team can iteratively deliver valuable functionality while ensuring quality, stakeholder satisfaction, and continuous improvement throughout the Agile software development lifecycle.